

The expected number of faces of a random embedding of a dense simple graph is $\Theta(\log n)$

Machine-generated note (see provenance box)

9 September 2026

Provenance

The mathematics in this note was produced without human intervention, in three stages. (1) The argument was found by the language model `gpt-5.6-sol` in a single autonomous pass on 2026-08-31, as part of a large sweep over open problems catalogued from arXiv (catalog id 2211.01032__02; original writeup in `attacks/2211.01032__02/output.md`). (2) An independent adversarial referee agent (`claude-fable-5`) re-derived every step, checked the source paper's $\text{\TeX}$ source, verified the exposure lemma, the facial-cycle probability and the light-face bound *exactly* by enumerating all rotation systems of K_5 and of the octahedron, and ran Monte Carlo experiments up to $n = 800$; its verdict was `MINOR_GAPS`: no substantive error, but a one-unit miscount in one intermediate display, implicit large- n thresholds, an unstated connectivity convention, and one unused inequality. Its report is reproduced verbatim in [Appendix B](#). (3) On 2026-09-09 the writeup was rewritten into the present note by `claude-fable-5.1`, repairing exactly those four points: the count of admissible darts in the heavy-face bound is corrected to $d_u - \lfloor \theta d_u \rfloor - 1$ ([Lemma 3.2](#); the downstream bound $2/\delta$ is unchanged); the theorem is stated for $n \geq n_0(c)$ and the thresholds used are listed ([Remark 6.1](#)); the connectivity convention is discussed ([Remark 6.2](#)); and the unused inequality is dropped. The text was reorganised with full proofs in [Appendix A](#). No other mathematical content was changed. **No human mathematician has checked this argument.** We circulate it to obtain exactly that: is the proof correct, and is the result new?

Abstract

A random embedding of a graph G is obtained by choosing, independently and uniformly at each vertex, a cyclic order of the incident edge-ends; let $F(G)$ be the resulting number of faces. Campion Loth, Halasz, Masařik, Mohar and Šámal conjectured (Conjecture 9.3 of arXiv:2211.01032) that every graph on n vertices with minimum degree $\Omega(n)$ has $\mathbb{E}F(G) = \Theta(\log n)$. We prove this for simple graphs: for every $c > 0$ and every simple graph G on $n \geq n_0(c)$ vertices with $\delta(G) \geq cn$,

$$\frac{1}{2} \log n - O_c(1) \leq \mathbb{E}F(G) \leq (2/c^2 + o_c(1)) \log n.$$

The upper bound exposes the random rotations along a face and splits faces into light, heavy-and-short, and long; the lower bound counts faces bounded by vertex-simple cycles through a spectral estimate. For K_n the upper bound specialises to $(2 + o(1)) \log n$. We also observe that the statement is false for multigraphs, so the simple-graph reading is the intended one.

1 Introduction

Random embeddings. Let $G = (V, E)$ be a finite graph, $n = |V|$, $m = |E|$ and $\delta = \delta(G)$ its minimum degree. Every edge consists of two *darts* (edge-ends), one based at each endpoint; let D be the set of $N := 2m$ darts, d_v the degree of v , and $\iota: D \rightarrow D$ the involution exchanging the two darts of every edge. A *rotation system* is a permutation $\rho = \prod_{v \in V} \rho_v$ of D in which each ρ_v is a cyclic permutation (d_v -cycle) of the darts based at v . It determines a cellular embedding of G in an orientable surface whose faces are the cycles of the *facial permutation* $\Phi = \rho\iota$ (the composition convention is immaterial for the number of cycles). The *random embedding* of G chooses each ρ_v uniformly and

independently among the $(d_v - 1)!$ cyclic permutations, and $F(G)$ denotes the number of faces, i.e. the number of cycles of Φ .

Campion Loth, Halasz, Masařík, Mohar and Šámal [1] proved $\frac{1}{2} \ln n - 2 < \mathbb{E}F(K_n) \leq 3.65 \ln n$ (for large n) and, observing that all known examples with $\mathbb{E}F = \Theta(n)$ have bounded degree, posed in their Section 9 (“Open Problems”):

Conjecture 1.1 ([1, Conjecture 9.3]). Let G be a graph on n vertices with minimum vertex degree $\Omega(n)$. Then G satisfies $\mathbb{E}[F] = \Theta(\log(n))$.

Conventions. The paper [1] says “multigraph” when multigraphs are meant and states that it is “mostly interested in simple graphs”; the referee confirmed this against the $\text{\TeX}$ source (Appendix B). We read Conjecture 1.1 for finite simple graphs. This is the only tenable reading: Proposition 5.1 below shows that for multigraphs the statement fails badly. All logarithms are natural; $H_N = \sum_{i \leq N} 1/i$ is the harmonic number.

Theorem 1.2 (Main theorem). For every $c > 0$ there are $n_0(c)$, a constant A_c and a function $\varepsilon_c(n) \rightarrow 0$ such that every simple graph G on $n \geq n_0(c)$ vertices with $\delta(G) \geq cn$ satisfies

$$\frac{1}{2} \log n - A_c \leq \mathbb{E}F(G) \leq \left(\frac{2}{c^2} + \varepsilon_c(n) \right) \log n.$$

More precisely, with $\theta = (\log n)^{-1/2}$, $T = \lfloor \theta \delta^2 / 100 \rfloor$ and $\gamma = \frac{1}{2} \ln(25/e)$, the upper bound

$$\mathbb{E}F(G) \leq \frac{N H_N}{(1 - \theta)^2 \delta^2} + \frac{N}{T} + N n e^{-\gamma \theta \delta} \quad (1)$$

holds for every simple graph with $\delta \geq 2$, $\theta \delta \geq 2$ and $(\frac{1}{2} - \theta) \delta \geq 1$. In particular $\mathbb{E}F(K_n) \leq (2 + o(1)) \log n$.

Corollary 1.3. Conjecture 1.1 holds for simple graphs, with Θ -constants depending on the constant c in the hypothesis $\delta(G) \geq cn$.

Idea of proof. Upper bound. Follow the facial orbit of a fixed dart a , revealing values of the random rotations only when they are queried. Until the face closes every queried input is new, so each step is a uniform choice among the not-yet-excluded darts at the current vertex (Lemma 2.1). Call a face *light* if it uses at most a θ -fraction of the darts at every vertex. Closing a light face of any length ℓ requires two specific choices, each of probability at most $1/((1 - \theta)\delta)$, so $\Pr(L(a) = \ell, \text{light}) \leq ((1 - \theta)\delta)^{-2}$ uniformly in ℓ (Lemma 3.1); with the rooting identity $F = \sum_a 1/L(a)$ this gives $\mathbb{E}F_{\text{light}} \leq N H_N / ((1 - \theta)^2 \delta^2)$. Heavy faces of length at most T are exponentially unlikely because a vertex must be selected $\Omega(\theta \delta)$ times at probability $\leq 2/\delta$ each (Lemma 3.2), and there are at most N/T faces longer than T . *Lower bound.* A rooted oriented simple ℓ -cycle is facial with probability exactly $\prod_i (d_{v_i} - 1)^{-1}$ (Lemma 4.1). Summing over cycles is a trace of a power of the normalised adjacency matrix $S = D^{-1/2} A D^{-1/2}$, up to the mass of non-simple closed walks, which is $O(\ell^2 n / \delta^2)$ (Lemma 4.2). Pairing $\text{tr } S^{2k} + \text{tr } S^{2k+1} \geq 2$ (the eigenvalue 1) and summing k up to $\Theta(\delta/\sqrt{n})$ gives $\frac{1}{2} \log n - O_c(1)$ (Lemma 4.3). [full proof]

2 The exposure lemma

Lemma 2.1 (Deferred decisions). Let π be a uniformly random cyclic permutation of a d -element set. Suppose $k \leq d - 2$ values of π have been exposed and the exposed arcs are extendible to a d -cycle. If x is a point whose image is not yet exposed, then conditionally on the exposed values, $\pi(x)$ is uniform among $d - k - 1$ admissible points.

Idea of proof. The exposed arcs form $d - k$ vertex-disjoint directed paths (isolated points allowed); x is the end of one of them, and $\pi(x)$ can be the start of any of the other $d - k - 1$ paths. Each choice leaves $d - k - 1$ paths to be closed into a cycle, in $(d - k - 2)!$ ways, so all choices are equiprobable. [\[full proof\]](#)

While the facial orbit $a, \Phi(a), \Phi^2(a), \dots$ of a dart a is followed, the inputs $\iota(\Phi^t(a))$ queried from the rotations are pairwise distinct until the orbit closes, because the darts of an orbit are distinct before the first return. Hence [Lemma 2.1](#) applies at every step of the exploration of one face. Simplicity of G enters through one fact: among the darts based at a vertex v , at most one leads to any prescribed neighbour w .

3 Upper bound

Throughout this section G is simple, $\theta = (\log n)^{-1/2}$, and we assume

$$\delta \geq 2, \quad \theta\delta \geq 2, \quad \left(\frac{1}{2} - \theta\right)\delta \geq 1 \quad (2)$$

(all of which hold for $n \geq n_0(c)$ when $\delta \geq cn$). For a face C and a vertex v let $r_C(v)$ be the number of darts of C based at v (the number of *transitions* of C at v), so $|C| = \sum_v r_C(v)$. The face C is *light* if $r_C(v) \leq \theta d_v$ for every v , and *heavy* otherwise. For a dart a let $C(a)$ be its face and $L(a) = |C(a)|$ its length.

Lemma 3.1 (Light faces). *For every dart a and every $\ell \geq 3$,*

$$\Pr(L(a) = \ell \text{ and } C(a) \text{ is light}) \leq \frac{1}{(1 - \theta)^2 \delta^2}.$$

Moreover, no face has length 1 or 2, and $\mathbb{E}F_{\text{light}} \leq \frac{N H_N}{(1 - \theta)^2 \delta^2}$, where F_{light} is the number of light faces.

Idea of proof. Decompose according to the history of the first $\ell - 2$ steps of the orbit. For the face to close at length ℓ , the penultimate choice (at a vertex v) must pick the unique dart leading to the base s of a , and the last choice (at s) must pick a itself. If the closed face is light, at most $r_C(v) - 1 \leq \theta d_v - 1$ values of ρ_v were exposed before the penultimate step, so by [Lemma 2.1](#) it is a uniform choice among at least $d_v - r_C(v) \geq (1 - \theta)\delta$ darts; likewise at s . The rooting identity $F = \sum_{a \in D} 1/L(a)$ and $\sum_{\ell \leq N} 1/\ell \leq H_N$ finish. [\[full proof\]](#)

Lemma 3.2 (Heavy faces are long). *Let $T = \lfloor \theta\delta^2/100 \rfloor$ and $\gamma = \frac{1}{2} \ln(25/e) > 0$. For every dart a ,*

$$\Pr(C(a) \text{ is heavy and } L(a) \leq T) \leq n e^{-\gamma\theta\delta},$$

and consequently $\mathbb{E}F_{\text{heavy,short}} \leq N n e^{-\gamma\theta\delta}$, where $F_{\text{heavy,short}}$ is the number of heavy faces of length at most T .

Idea of proof. Explore the orbit of a and stop as soon as some vertex v has been queried $\lfloor \theta d_v \rfloor + 1$ times. Before the stopping time, a query at u has at most $\lfloor \theta d_u \rfloor$ exposed values, so by [Lemma 2.1](#) it is uniform over at least $d_u - \lfloor \theta d_u \rfloor - 1 \geq \delta/2$ darts (this is the corrected count; the original writeup omitted the -1), of which at most one leads to a prescribed vertex w : the next vertex is w with conditional probability at most $2/\delta$. A heavy face of length $\leq T$ forces some w to be selected at least $K_w = \lfloor \theta d_w \rfloor \geq \theta\delta/2$ times within T steps, which has probability at most $\binom{T}{K_w} (2/\delta)^{K_w} \leq (e/25)^{K_w} \leq e^{-\gamma\theta\delta}$; take a union bound over w . [\[full proof\]](#)

Lemma 3.3 (Few long faces). *Deterministically, the number F_{long} of faces of length greater than T is at most N/T .*

Proof. The faces partition the N darts. □

4 Lower bound

Let A be the adjacency matrix of G , $D = \text{diag}(d_v)$, $S = D^{-1/2}AD^{-1/2}$ and $P = D^{-1}A$. Let X_ℓ be the number of faces whose boundary is a vertex-simple cycle of length ℓ .

Lemma 4.1 (Facial simple cycles). *For every rooted oriented simple cycle $(v_0, v_1, \dots, v_{\ell-1}, v_0)$ of G with $\ell \geq 3$, the probability that it is a facial cycle is exactly $\prod_{i=0}^{\ell-1} (d_{v_i} - 1)^{-1}$. Consequently*

$$\mathbb{E}X_\ell \geq \frac{W_\ell}{\ell}, \quad W_\ell := \sum_{\substack{v_0 \cdots v_{\ell-1} v_0 \text{ closed walk} \\ v_0, \dots, v_{\ell-1} \text{ distinct}}} \prod_{i=0}^{\ell-1} \frac{1}{d_{v_i}}.$$

Idea of proof. The cycle is facial iff at every v_i the rotation ρ_{v_i} sends the dart towards v_{i-1} to the dart towards v_{i+1} ; these are two distinct darts (the cycle is simple), the successor of a fixed dart in a uniform d -cycle is uniform over the other $d - 1$ darts, and the rotations are independent. Each such face is counted by ℓ rooted oriented cycles, and $1/(d - 1) \geq 1/d$. [\[full proof\]](#)

Lemma 4.2 (Non-simple closed walks). *$\text{tr } S^\ell = \sum \prod_{i=0}^{\ell-1} d_{v_i}^{-1}$, the sum over all rooted closed walks $v_0 v_1 \cdots v_{\ell-1} v_0$ of length ℓ , and for every $\ell \geq 3$*

$$W_\ell \geq \text{tr } S^\ell - B_\ell, \quad B_\ell := \binom{\ell}{2} \frac{n}{\delta^2}.$$

Idea of proof. $(S^r)_{vv} = P^r(v, v) \leq \max_u P(u, v) \leq 1/\delta$ for $r \geq 1$, using simplicity. A closed walk with $v_i = v_j$ ($i < j$) splits at that vertex into two closed walks, so the weight of such walks is $\sum_v (S^r)_{vv} (S^{\ell-r})_{vv} \leq n/\delta^2$; a union bound over the $\binom{\ell}{2}$ pairs of positions gives the claim. [\[full proof\]](#)

Lemma 4.3 (Pairing even and odd lengths). *For every $k \geq 1$, $\text{tr } S^{2k} + \text{tr } S^{2k+1} \geq 2$, and $\frac{\text{tr } S^{2k}}{2k} + \frac{\text{tr } S^{2k+1}}{2k+1} \geq \frac{2}{2k+1}$.*

Idea of proof. The eigenvalues of S lie in $[-1, 1]$, so every term $\lambda_j^{2k}(1 + \lambda_j)$ is nonnegative, and each connected component contributes an eigenvalue 1 with term 2. Since $\text{tr } S^{2k} \geq 0$, the weighted sum is at least $(\text{tr } S^{2k} + \text{tr } S^{2k+1})/(2k + 1)$. [\[full proof\]](#)

5 The multigraph reading is false

Proposition 5.1. *For every even n there is a connected multigraph on n vertices with minimum degree at least n for which every rotation system has at least $n/2 + 1$ faces. Hence [Conjecture 1.1](#) fails for multigraphs.*

Idea of proof. Chain $k = n/2$ dipoles (two vertices joined by $M = n$ parallel edges) by single bridge edges. A dipole with M edges has $F = M - 2g$ faces in any orientable embedding, hence $F \geq 2$ as M is even; a bridge merges exactly two faces into one, so the chain has $F \geq 2k - (k - 1) = k + 1$. [\[full proof\]](#)

6 Remarks

Remark 6.1 (Thresholds). The proof uses only the following consequences of $\delta \geq cn$ and $n \geq n_0(c)$, with $\theta = (\log n)^{-1/2}$: $\delta \geq 2$ (no faces of length ≤ 2); $(\frac{1}{2} - \theta)\delta \geq 1$ (so that $d_u - \lfloor \theta d_u \rfloor - 1 \geq \delta/2$ in [Lemma 3.2](#), and $\lfloor \theta d_u \rfloor \leq d_u - 2$ so that [Lemma 2.1](#) applies); $\theta\delta \geq 2$ (so that $K_w = \lfloor \theta d_w \rfloor \geq \theta\delta/2$); $T \geq \theta\delta^2/200$; and $K = \lfloor \delta/(4\sqrt{n}) \rfloor \geq 2$ in the lower bound. The original writeup left these implicit. Since the statement is a $\Theta(\log n)$ statement, restricting to $n \geq n_0(c)$ is harmless.

Remark 6.2 (Connectivity). The source paper defines random embeddings for connected graphs; a graph with $\delta \geq cn$ may have up to $\lfloor 1/c \rfloor$ components. Both bounds hold verbatim for disconnected G , with F the total number of faces over the components: the upper bound is proved dart by dart and never uses connectivity, and in the lower bound every component contributes an eigenvalue 1 of S , which is all that [Lemma 4.3](#) needs (it only uses that at least one such eigenvalue exists). So [Theorem 1.2](#) covers both the connected and the general reading.

Remark 6.3 (Dependence on c). The upper constant $2/c^2$ blows up as $c \rightarrow 0$; this is inherent in the hypothesis $\delta = \Omega(n)$ and consistent with the conjecture's Θ . For K_n , $N/\delta^2 = n/(n-1)$ and (1) gives $\mathbb{E}F(K_n) \leq (2 + o(1)) \log n$. The referee observed that this matches the constant in the Mauk–Stahl conjecture $\mathbb{E}F(K_n) = 2 \ln n + O(1)$ restated in [1, Section 9], and improves the constant 3.65 of [1, Theorem 1.8]; likewise for dense $G(n, p)$, where $N/\delta^2 \approx 1/p$, (1) gives $O_p(\log n)$. We record these as consequences the referee noted, not as independently checked claims.

Remark 6.4 (Computational checks by the referee). By enumerating all 7776 rotation systems of K_5 and all 46656 of the octahedron, the referee verified exactly the rooting identity, the facial-cycle probability of [Lemma 4.1](#) ($1/27, 1/81, 1/243$), the bound of [Lemma 3.1](#) (which is exactly tight for $\theta = 1/4$ on K_5) and the inequality $\mathbb{E}X_\ell \geq (\text{tr } S^\ell - B_\ell)/\ell$; [Lemma 2.1](#) was verified on all 120 cyclic permutations of six points; Monte Carlo on K_n up to $n = 800$ gave $\mathbb{E}F(K_n) \approx 2 \ln n - 0.9$, below the upper bound $2 \ln n + O(\sqrt{\log n})$ and far above the lower bound $\frac{1}{2} \ln n$; and the dipole chain of [Proposition 5.1](#) was checked exactly. Scripts: `verification/scripts/2211.01032__02/`.

Remark 6.5 (Novelty and a caveat). The referee's search found no resolution of [Conjecture 1.1](#) in the literature. It also noted that a four-page elementary argument improving the constants of a long SODA paper is a reason for caution, and recommended an independent expert pass before the conjecture is treated as resolved; we repeat that recommendation. Nothing here has been checked by a human.

A Full proofs

Lemma 2.1 (Deferred decisions). *Let π be a uniformly random cyclic permutation of a d -element set. Suppose $k \leq d - 2$ values of π have been exposed and the exposed arcs are extendible to a d -cycle. If x is a point whose image is not yet exposed, then conditionally on the exposed values, $\pi(x)$ is uniform among $d - k - 1$ admissible points.*

Proof. Condition on the k exposed values. They form a set of arcs $y \mapsto \pi(y)$ which, being part of a single d -cycle, consists of vertex-disjoint directed paths; counting isolated points as paths of length 0, there are exactly $d - k$ such paths. The point x is the terminal point of one of them, say Q (possibly $Q = \{x\}$). A d -cycle extending the exposed arcs is obtained by arranging the $d - k$ paths in a cyclic order and joining the end of each path to the start of the next; distinct cyclic orders give distinct cycles and every extension arises this way, so there are $(d - k - 1)!$ extensions, all equally likely under the uniform distribution on d -cycles. The image $\pi(x)$ is the start of the path following Q in the cyclic order; it can be any of the other $d - k - 1$ paths (there is at least one since $k \leq d - 2$), and for each choice the remaining $d - k - 1$ blocks can be cyclically ordered in $(d - k - 2)!$ ways. Hence each of the $d - k - 1$ admissible values of $\pi(x)$ has conditional probability $(d - k - 2)!/(d - k - 1)! = 1/(d - k - 1)$. $\square$

Lemma 3.1 (Light faces). *For every dart a and every $\ell \geq 3$,*

$$\Pr(L(a) = \ell \text{ and } C(a) \text{ is light}) \leq \frac{1}{(1 - \theta)^2 \delta^2}.$$

Moreover, no face has length 1 or 2, and $\mathbb{E}F_{\text{light}} \leq \frac{NH_N}{(1 - \theta)^2 \delta^2}$, where F_{light} is the number of light faces.

Proof. No short faces. A face of length 1 is a dart a with $\rho(\iota(a)) = a$, which requires $\iota(a)$ and a to be based at the same vertex, i.e. a loop. A face of length 2 consists of darts a (based at u , towards v) and $b = \rho_v(\iota(a))$ with $\rho(\iota(b)) = a$; the latter forces $\iota(b)$ to be based at u , so b is a dart from v to u , and in a simple graph $b = \iota(a)$. Then $\rho_v(\iota(a)) = \iota(a)$ is a fixed point of the d_v -cycle ρ_v , impossible for $d_v \geq 2$.

The probability bound. Fix a , based at s , and $\ell \geq 3$. Expose the orbit $x_0 = a$, $x_{t+1} = \rho(\iota(x_t))$ step by step; by the remark after Lemma 2.1, every step until the orbit closes queries a fresh input of the relevant rotation, so Lemma 2.1 applies (the number of exposed values at a vertex v before a query at v equals the number of earlier transitions of the orbit at v). Let h range over the possible histories $(x_0, \dots, x_{\ell-2})$ of the first $\ell - 2$ steps in which the orbit has not yet closed. Given h , the orbit closes at length exactly ℓ only if the next two steps are forced: with v the vertex at which $\iota(x_{\ell-2})$ is based, the query at v must return the dart $x_{\ell-1}$ from v to s (there is at most one such dart, as G is simple; and $v \neq s$, since $v = s$ would make $x_{\ell-2}$ a loop), and then the query at s must return $x_\ell = a$. Thus given h the face C obtained on closing is determined, and so is whether it is light. Let $\mathcal{H}$ be the set of histories h for which this face is light and both forced darts exist. Then

$$\Pr(L(a) = \ell, C(a) \text{ light}) = \sum_{h \in \mathcal{H}} \Pr(h) \Pr(x_{\ell-1} \text{ forced} \mid h) \Pr(x_\ell = a \mid h, x_{\ell-1}).$$

Fix $h \in \mathcal{H}$ with its light face C . The transitions of C at v are its darts based at v ; the forced dart $x_{\ell-1}$ is one of them, so at most $r_C(v) - 1 \leq \lfloor \theta d_v \rfloor - 1 \leq d_v - 2$ values of ρ_v have been exposed before the query at v , and by Lemma 2.1 the query is uniform over at least $d_v - (r_C(v) - 1) - 1 = d_v - r_C(v) \geq (1 - \theta)d_v \geq (1 - \theta)\delta$ admissible darts. Hence $\Pr(x_{\ell-1} \text{ forced} \mid h) \leq 1/((1 - \theta)\delta)$ (and $= 0$ if the forced dart is inadmissible). Likewise the darts of C based at s include a itself, which is produced by the final query, so before the final query at most $r_C(s) - 1$ values of ρ_s have been exposed, and $\Pr(x_\ell = a \mid h, x_{\ell-1}) \leq 1/((1 - \theta)\delta)$. Summing over $h \in \mathcal{H}$ and using $\sum_h \Pr(h) \leq 1$ gives the bound.

Expectation. Every face of length ℓ contains exactly ℓ darts, so $F = \sum_{a \in D} 1/L(a)$ and $F_{\text{light}} = \sum_{a \in D} \mathbf{1}[C(a) \text{ light}]/L(a)$. As $3 \leq L(a) \leq N$,

$$\mathbb{E}F_{\text{light}} = \sum_{a \in D} \sum_{\ell=3}^N \frac{1}{\ell} \Pr(L(a) = \ell, C(a) \text{ light}) \leq \frac{N}{(1 - \theta)^2 \delta^2} \sum_{\ell=3}^N \frac{1}{\ell} \leq \frac{N H_N}{(1 - \theta)^2 \delta^2}. \quad \square$$

Lemma 3.2 (Heavy faces are long). *Let $T = \lfloor \theta \delta^2 / 100 \rfloor$ and $\gamma = \frac{1}{2} \ln(25/e) > 0$. For every dart a ,*

$$\Pr(C(a) \text{ is heavy and } L(a) \leq T) \leq n e^{-\gamma \theta \delta},$$

and consequently $\mathbb{E}F_{\text{heavy, short}} \leq N n e^{-\gamma \theta \delta}$, where $F_{\text{heavy, short}}$ is the number of heavy faces of length at most T .

Proof. Fix a and explore its orbit as above. Let τ be the first step at which some vertex v has been queried $\lfloor \theta d_v \rfloor + 1$ times (the query at the base of a producing a itself is not counted as a query). Consider a query at a vertex u made at a step $t \leq \tau$. Before it, at most $\lfloor \theta d_u \rfloor$ values of ρ_u were exposed (if the count were larger, τ would have occurred earlier), and $\lfloor \theta d_u \rfloor \leq d_u - 2$ by (2). By Lemma 2.1 the query is uniform over at least

$$d_u - \lfloor \theta d_u \rfloor - 1 \geq (1 - \theta)d_u - 1 \geq (1 - \theta)\delta - 1 \geq \frac{\delta}{2}$$

admissible darts, where the last step is $(\frac{1}{2} - \theta)\delta \geq 1$ from (2). (The original writeup wrote $d_u - \lfloor \theta d_u \rfloor$ here, overcounting by one; the conclusion below is unaffected.) Since G is simple, at most one admissible dart leads to any prescribed vertex w , so for every w and every step $t \leq \tau$,

$$\Pr(\text{the query at step } t \text{ selects } w \mid \text{past}) \leq \frac{2}{\delta}. \quad (3)$$

Suppose $C(a)$ is heavy and $L(a) \leq T$. Then some vertex w has $r_C(w) \geq \lfloor \theta d_w \rfloor + 1$ transitions, so $\tau \leq L(a) \leq T$ occurs, at some vertex w^* that was queried $\lfloor \theta d_{w^*} \rfloor + 1$ times within the first T steps. At most one of these queries is the initial one (if w^* is the base of a), so w^* was *selected* (chosen as the head of the dart returned by a query) at least $K_{w^*} := \lfloor \theta d_{w^*} \rfloor$ times at steps $\leq \tau$. For a fixed vertex w and a fixed set of K_w steps, iterating (3) shows that the probability that all these steps select w (and are $\leq \tau$) is at most $(2/\delta)^{K_w}$; a union bound over the $\binom{T}{K_w}$ sets of steps gives

$$\Pr(w \text{ is selected at least } K_w \text{ times at steps } \leq \min\{\tau, T\}) \leq \binom{T}{K_w} \left(\frac{2}{\delta}\right)^{K_w} \leq \left(\frac{2eT}{\delta K_w}\right)^{K_w}.$$

By (2), $K_w \geq \theta d_w - 1 \geq \theta\delta - 1 \geq \theta\delta/2$, and $T \leq \theta\delta^2/100$, so $2eT/(\delta K_w) \leq 2e \cdot \frac{\theta\delta^2/100}{\delta \cdot \theta\delta/2} = e/25 < 1$. Hence the probability is at most $(e/25)^{K_w} \leq (e/25)^{\theta\delta/2} = e^{-\gamma\theta\delta}$ with $\gamma = \frac{1}{2} \ln(25/e)$. (If $K_w > T$ the probability is 0.) A union bound over the n vertices w gives $\Pr(C(a) \text{ heavy}, L(a) \leq T) \leq ne^{-\gamma\theta\delta}$. Finally $F_{\text{heavy}, \text{short}} = \sum_a \mathbf{1}[C(a) \text{ heavy}, L(a) \leq T]/L(a) \leq \sum_a \mathbf{1}[C(a) \text{ heavy}, L(a) \leq T]$, so $\mathbb{E}F_{\text{heavy}, \text{short}} \leq Nne^{-\gamma\theta\delta}$. $\square$

Lemma 4.1 (Facial simple cycles). *For every rooted oriented simple cycle $(v_0, v_1, \dots, v_{\ell-1}, v_0)$ of G with $\ell \geq 3$, the probability that it is a facial cycle is exactly $\prod_{i=0}^{\ell-1} (d_{v_i} - 1)^{-1}$. Consequently*

$$\mathbb{E}X_\ell \geq \frac{W_\ell}{\ell}, \quad W_\ell := \sum_{\substack{v_0 \dots v_{\ell-1} v_0 \text{ closed walk} \\ v_0, \dots, v_{\ell-1} \text{ distinct}}} \prod_{i=0}^{\ell-1} \frac{1}{d_{v_i}}.$$

Proof. Let a_i be the dart from v_i to v_{i+1} (indices modulo ℓ); these darts exist and are distinct since the v_i are distinct and consecutive ones adjacent. The cycle is facial iff $\Phi(a_i) = a_{i+1}$ for all i , i.e. $\rho_{v_{i+1}}(\iota(a_i)) = a_{i+1}$: the rotation at v_{i+1} sends the dart towards v_i to the dart towards v_{i+2} . These two darts are distinct because $v_i \neq v_{i+2}$ (as $\ell \geq 3$). For a uniformly random d -cycle the successor of a fixed point is uniform over the other $d-1$ points, so the condition at v_{i+1} has probability $1/(d_{v_{i+1}} - 1)$; the ℓ conditions concern the ℓ distinct vertices $v_0, \dots, v_{\ell-1}$, whose rotations are independent. Hence the probability is $\prod_i (d_{v_i} - 1)^{-1}$.

A face whose boundary is a vertex-simple ℓ -cycle traverses its darts in a definite cyclic direction, so it corresponds to exactly ℓ rooted oriented simple cycles (the choice of the root). Therefore

$$\mathbb{E}X_\ell = \frac{1}{\ell} \sum_{(v_0, \dots, v_{\ell-1})} \prod_i \frac{1}{d_{v_i} - 1} \geq \frac{1}{\ell} \sum_{(v_0, \dots, v_{\ell-1})} \prod_i \frac{1}{d_{v_i}} = \frac{W_\ell}{\ell},$$

the sums running over rooted oriented simple ℓ -cycles, which are exactly the closed walks of length ℓ with distinct $v_0, \dots, v_{\ell-1}$. $\square$

Lemma 4.2 (Non-simple closed walks). *$\text{tr } S^\ell = \sum \prod_{i=0}^{\ell-1} d_{v_i}^{-1}$, the sum over all rooted closed walks $v_0 v_1 \dots v_{\ell-1} v_0$ of length ℓ , and for every $\ell \geq 3$*

$$W_\ell \geq \text{tr } S^\ell - B_\ell, \quad B_\ell := \binom{\ell}{2} \frac{n}{\delta^2}.$$

Proof. $S_{uv} = A_{uv}/\sqrt{d_u d_v}$, so $(S^\ell)_{v_0 v_0} = \sum_{v_1, \dots, v_{\ell-1}} \prod_i A_{v_i v_{i+1}}/\sqrt{d_{v_i} d_{v_{i+1}}}$ (indices mod ℓ), and each vertex position v_i appears in exactly two square roots, giving the factor $1/d_{v_i}$; summing over v_0 gives the formula for $\text{tr } S^\ell$. Since $S = D^{1/2} P D^{-1/2}$, $(S^r)_{vv} = P^r(v, v)$, and for $r \geq 1$

$$P^r(v, v) = \sum_u P^{r-1}(v, u) P(u, v) \leq \max_u P(u, v) \leq \frac{1}{\delta},$$

because P^{r-1} is row-stochastic and $P(u, v) = A_{uv}/d_u \leq 1/\delta$ in a simple graph (for $r = 1$, $P(v, v) = 0$).

Fix positions $0 \leq i < j \leq \ell - 1$ and let $r = j - i$. A closed walk $v_0 \cdots v_{\ell-1} v_0$ with $v_i = v_j =: v$ splits into the closed walk $v_i v_{i+1} \cdots v_j$ of length r at v and the closed walk $v_j v_{j+1} \cdots v_{\ell-1} v_0 \cdots v_i$ of length $\ell - r$ at v , and its weight $\prod_{t=0}^{\ell-1} d_{v_t}^{-1}$ is the product of the weights of the two pieces (positions $i, \dots, j-1$ and $j, \dots, \ell-1, 0, \dots, i-1$). Hence the total weight of closed walks with $v_i = v_j$ is $\sum_v (S^r)_{vv} (S^{\ell-r})_{vv} \leq n/\delta^2$; if $r = 1$ or $\ell - r = 1$ it is 0, as G has no loops. Every closed walk with a repeated vertex among $v_0, \dots, v_{\ell-1}$ has $v_i = v_j$ for some pair, and all weights are nonnegative, so the weight of non-simple closed walks is at most $\binom{\ell}{2} n/\delta^2 = B_\ell$, and $W_\ell \geq \text{tr } S^\ell - B_\ell$. $\square$

Lemma 4.3 (Pairing even and odd lengths). *For every $k \geq 1$, $\text{tr } S^{2k} + \text{tr } S^{2k+1} \geq 2$, and $\frac{\text{tr } S^{2k}}{2k} + \frac{\text{tr } S^{2k+1}}{2k+1} \geq \frac{2}{2k+1}$.*

Proof. S is symmetric and similar to the row-stochastic matrix P , so its eigenvalues λ_j are real and lie in $[-1, 1]$. For each connected component H of G , the vector $D^{1/2} \mathbf{1}_H$ satisfies $SD^{1/2} \mathbf{1}_H = D^{-1/2} A \mathbf{1}_H = D^{1/2} \mathbf{1}_H$, so 1 is an eigenvalue (with multiplicity at least the number of components). Then $\text{tr } S^{2k} + \text{tr } S^{2k+1} = \sum_j \lambda_j^{2k} (1 + \lambda_j)$, every term is nonnegative, and the eigenvalue 1 contributes 2; hence the sum is at least 2. For the second claim, $\text{tr } S^{2k} = \sum_j \lambda_j^{2k} \geq 0$ gives $\text{tr } S^{2k}/(2k) \geq \text{tr } S^{2k}/(2k+1)$, so the left side is at least $(\text{tr } S^{2k} + \text{tr } S^{2k+1})/(2k+1) \geq 2/(2k+1)$. $\square$

Theorem 1.2 (Main theorem). *For every $c > 0$ there are $n_0(c)$, a constant A_c and a function $\varepsilon_c(n) \rightarrow 0$ such that every simple graph G on $n \geq n_0(c)$ vertices with $\delta(G) \geq cn$ satisfies*

$$\frac{1}{2} \log n - A_c \leq \mathbb{E}F(G) \leq \left(\frac{2}{c^2} + \varepsilon_c(n) \right) \log n.$$

More precisely, with $\theta = (\log n)^{-1/2}$, $T = \lfloor \theta \delta^2 / 100 \rfloor$ and $\gamma = \frac{1}{2} \ln(25/e)$, the upper bound

$$\mathbb{E}F(G) \leq \frac{NH_N}{(1-\theta)^2 \delta^2} + \frac{N}{T} + Nn e^{-\gamma \theta \delta} \quad (1)$$

holds for every simple graph with $\delta \geq 2$, $\theta \delta \geq 2$ and $(\frac{1}{2} - \theta) \delta \geq 1$. In particular $\mathbb{E}F(K_n) \leq (2 + o(1)) \log n$.

Proof. Let G be simple with $\delta \geq cn$ and $n \geq n_0(c)$, where $n_0(c)$ is large enough that (2) holds, that $T \geq \theta \delta^2 / 200$, and that $K := \lfloor \delta / (4\sqrt{n}) \rfloor \geq 2$.

Upper bound. Every face is light, or heavy of length at most T , or of length greater than T , so $F \leq F_{\text{light}} + F_{\text{heavy, short}} + F_{\text{long}}$. Lemmas 3.1 to 3.3 give (1):

$$\mathbb{E}F(G) \leq \frac{NH_N}{(1-\theta)^2 \delta^2} + \frac{N}{T} + Nn e^{-\gamma \theta \delta}.$$

Now $N = 2m \leq n^2$, $H_N \leq \ln N + 1 \leq 2 \log n + 1$, and $N/\delta^2 \leq n^2/(cn)^2 = 1/c^2$, so the first term is at most $(1-\theta)^{-2} (2 \log n + 1)/c^2 = (2/c^2) \log n (1 + O(\theta)) + O(1/c^2)$. Next $N/T \leq 200n^2/(\theta c^2 n^2) = 200\sqrt{\log n}/c^2 = o(\log n)$. Finally $Nn e^{-\gamma \theta \delta} \leq n^3 \exp(-\gamma cn/\sqrt{\log n}) = o(1)$. Hence $\mathbb{E}F(G) \leq (2/c^2 + \varepsilon_c(n)) \log n$ with $\varepsilon_c(n) \rightarrow 0$ depending only on c and n . For K_n , $N/\delta^2 = n(n-1)/(n-1)^2 = n/(n-1)$, and the same computation gives $\mathbb{E}F(K_n) \leq (2 + o(1)) \log n$.

Lower bound. Faces bounded by simple cycles of different lengths are distinct, and $2K + 1 \leq \delta/(2\sqrt{n}) + 1 \leq n$, so $F \geq \sum_{k=2}^K (X_{2k} + X_{2k+1})$. By Lemmas 4.1 and 4.2,

$$\mathbb{E}F(G) \geq \sum_{k=2}^K \left(\frac{\text{tr } S^{2k}}{2k} + \frac{\text{tr } S^{2k+1}}{2k+1} \right) - \sum_{k=2}^K \left(\frac{B_{2k}}{2k} + \frac{B_{2k+1}}{2k+1} \right).$$

By Lemma 4.3 the first sum is at least $\sum_{k=2}^K \frac{2}{2k+1} \geq \sum_{k=2}^K \frac{1}{k+1} \geq \ln(K+2) - \ln 3 \geq \log K - 2$. For the error sum, $B_\ell/\ell = (\ell-1)n/(2\delta^2)$, so $B_{2k}/(2k) + B_{2k+1}/(2k+1) = (4k-1)n/(2\delta^2) < 2kn/\delta^2$, and $\sum_{k=2}^K 2k \leq K^2 + K \leq 2K^2$; since $K \leq \delta/(4\sqrt{n})$, the error sum is at most $2nK^2/\delta^2 \leq 1/8$. Hence $\mathbb{E}F(G) \geq \log K - 3$. Finally $K \geq \delta/(4\sqrt{n}) - 1 \geq c\sqrt{n}/4 - 1 \geq c\sqrt{n}/8$ for $n \geq n_0(c)$, so $\log K \geq \frac{1}{2} \log n + \log(c/8)$ and $\mathbb{E}F(G) \geq \frac{1}{2} \log n - A_c$ with $A_c = 3 - \log(c/8)$. $\square$

Proposition 5.1. *For every even n there is a connected multigraph on n vertices with minimum degree at least n for which every rotation system has at least $n/2 + 1$ faces. Hence Conjecture 1.1 fails for multigraphs.*

Proof. Let $n = 2k$ and $M = n$. Take k disjoint dipoles $Q_1, \dots, Q_k$, each consisting of two vertices joined by M parallel edges, and add the $k-1$ bridge edges e_i joining a vertex of Q_i to a vertex of Q_{i+1} . The resulting multigraph G is connected, has n vertices, and every vertex has degree M or $M+1$ or $M+2$, so $\delta(G) \geq n$.

A rotation system of a graph with V vertices, E edges and F faces defines an orientable embedding of genus g with $V - E + F = 2 - 2g$. For a dipole, $V = 2$, $E = M$, so $F = M - 2g$; as M is even, F is even, and $F \geq 1$ gives $F \geq 2$. Now let ρ be any rotation system of G . Removing a bridge e from an embedded connected graph disconnects it, and the two darts of e lie on a single face (a face traversing a bridge must traverse it in both directions, since the bridge separates the graph); deleting e (i.e. removing its two darts from the rotations) turns that face into one face of each of the two components and leaves all other faces unchanged. Hence $F(G, \rho) = F(G - e, \rho|_{G-e}) - 1$. Applying this to the $k-1$ bridges in turn, $F(G, \rho) = \sum_{i=1}^k F(Q_i, \rho|_{Q_i}) - (k-1) \geq 2k - (k-1) = k+1 = n/2 + 1$. So $\mathbb{E}F(G) \geq n/2 + 1 = \Omega(n)$, not $O(\log n)$. $\square$

B Report of the LLM referee (verbatim)

The following is the unedited report of the adversarial referee agent (claude-fable-5, 2026-09-02) on the *original* writeup. Its section and equation numbers refer to that writeup, not to this note; the four items it lists as defects are the ones repaired here (see the provenance box). The scripts it mentions are in `verification/scripts/2211.01032__02/`. Treat it as machine output, not as an authority.

B.1 Interpretation

Verbatim source statement (checked against the arXiv v3 TeX source, Section 9 “Open Problems”):

Let G be a graph on n vertices with minimum vertex degree $\Omega(n)$. Then G satisfies $\mathbb{E}[F] = \Theta(\log(n))$.

surrounded by: “A large family of examples of graphs on n vertices with $\mathbb{E}[F] = \Theta(n)$ are given in [CHMMS22]. However all of these examples have maximum degree $O(1)$... We were unable to find any examples of dense graphs with such a large number of average faces, which leads us to the next conjecture.” This matches the catalog text exactly.

Simple vs multigraph. The paper uses “multigraph” explicitly when it means multigraphs (Theorems 1.11 vs 1.12) and says “However, we are mostly interested in simple graphs.” The writeup proves the conjecture for finite simple graphs and additionally shows (Section 5, verified computationally below) that the multigraph reading is trivially FALSE (a chain of n -edge dipoles joined by bridges is a connected multigraph with $\delta \geq n$ and deterministically $F \geq n/2 + 1$). Since the source authors certainly knew that dipoles/multistars produce many faces, the simple-graph reading is unquestionably the intended one. The writeup’s interpretation is the intended interpretation, and a reasonable author of the source paper would consider their conjecture resolved by the writeup’s theorem (two-sided

bounds $\frac{1}{2} \log n - O_c(1) \leq \mathbb{E}F \leq (2/c^2 + o_c(1)) \log n$ uniformly over simple G with $\delta(G) \geq cn$; c -dependence of the Θ -constants is inherent in an $\Omega(n)$ hypothesis).

Connectivity. The source defines random 2-cell embeddings for connected graphs. $\delta \geq cn$ permits up to $\lfloor 1/c \rfloor$ components; the writeup is silent on this, but its upper bound is per-dart and its lower bound uses “each connected component contributes an eigenvalue 1”, so both bounds hold verbatim for disconnected G (faces summed over components). Convention gap only, not a mathematical one.

B.2 Step-by-step findings

step	label	note
§1 model: $\Phi = \rho\iota$, $F = \#\text{cycles}(\Phi)$	VALID	Standard rotation-system face count. Verified computationally: exact $\mathbb{E}F(K_5) = 1631/648$, and the face-trace matches the permutation model on all 7776 (resp. 46656) rotation systems of K_5 (octahedron).
Lemma 2.1 (exposure lemma)	VALID	Re-derived: $d - k$ path components, $d - k - 1$ admissible images, each with $(d - k - 2)!$ completions. Verified by full enumeration of all 120 6-cycles under 5 exposure patterns: conditional image exactly uniform on exactly $d - k - 1$ points in every case.
§2 claim: queried inputs of each ρ_v all distinct until face closes	VALID	Inputs are $\iota(x_t)$ for distinct darts x_t of one Φ -orbit before first return; distinct darts give distinct inputs. Conditioning on the trace history = conditioning on exposed arcs of independent per-vertex rotations, so Lemma 2.1 applies at every step.
§2 simplicity claim (≤ 1 candidate dart to a prescribed neighbor)	VALID	Immediate for simple graphs.

step	label	note
Lemma 3.1	VALID	The subtle point is conditioning on lightness of the <i>final</i> face. Re-derived carefully: decompose over histories h of the first $\ell - 2$ steps compatible with closing to a light face; the penultimate transition is at $v = \text{head}(x_{\ell-2}) \neq s$ (equality would force a loop), and is v's last transition of C, so $\leq r_C(v) - 1 \leq \theta d_v - 1$ prior exposures at v; admissible $\geq d_v - r_C(v) \geq (1 - \theta)d_v$; likewise at s (the free initial dart a is counted in $r_C(s)$, so exposures $\leq r_C(s) - 1$). Inadmissible targets give probability 0. Hence $\Pr \leq ((1 - \theta)\delta)^{-2}$ uniformly in ℓ. Verified exactly on K_5/octahedron for $\theta \in \{0.25, 0.5, 0.6\}$ — the bound holds and is <i>exactly tight</i> at $\theta = 0.25$ on K_5 ($1/9$) — and by Monte Carlo on K_{60} ($\theta = 0.3$, 500k unbiased traces): observed $\Pr(L = \ell, \text{light}) \approx 3.0 \times 10^{-4}$ for all $\ell \leq 12$, below the bound 5.9×10^{-4}, and flat in ℓ as the lemma predicts.
§3.1: no faces of length ≤ 2 ; identity $F = \sum_a 1/L(a)$; (3.1)	VALID	Length-1 face needs a loop; length-2 face forces $\rho_v(\iota(a)) = \iota(a)$, a fixed point of a d_v-cycle with $d_v \geq 2$ — impossible. Rooting identity trivial (verified in enumeration). (3.1): $\sum_{\ell=3}^N \ell^{-1} \leq H_N$, giving $\mathbb{E}F_{\text{light}} \leq NH_N/((1 - \theta)^2\delta^2)$.

step	label	note
§3.2 display (3.2)	VALID (one-unit slip)	As written, “Lemma 2.1 leaves at least $d_u - \lfloor \theta d_u \rfloor$ candidate darts” overcounts by one: with $\leq \lfloor \theta d_u \rfloor$ prior exposures, Lemma 2.1 leaves $\geq d_u - \lfloor \theta d_u \rfloor - 1$. The operative conclusion $\Pr(\text{next query at } w \mid \text{past}) \leq 2/\delta$ still holds for all large n (need $(\frac{1}{2} - \theta)\delta \geq 1$, true since $\theta \rightarrow 0, \delta \geq cn$), and only $2/\delta$ is used downstream. Trivially repairable.
§3.2 heavy-count argument and (3.3)	VALID	Heavy at w with $L \leq T$ forces $\geq K_w = \lfloor \theta d_w \rfloor$ selections of w (initial dart exempted), all before the stopping time, each with adaptive probability $\leq 2/\delta$; fixed-time-set iterated conditioning justifies $\binom{T}{K_w}(2/\delta)^{K_w}$. Arithmetic re-derived: $2eT/(\delta K_w) \leq 2e \cdot \frac{\theta \delta^2/100}{\delta \cdot \theta \delta/2} = e/25 < 1$, so bound $\leq \exp(-\gamma \theta \delta)$ with $\gamma = \frac{1}{2} \ln(25/e) \approx 1.1$. ($K_w > T$ gives probability 0; $K_w \geq \theta \delta/2$ needs $\theta \delta \geq 2$, true for large n .)
(3.4), (3.5)	VALID	Union bound over $w \in V$; $1/L \leq 1$;
(3.6) long faces	VALID	$\mathbb{E}F_{\text{heavy,short}} \leq Nne^{-\gamma \theta \delta}$. Faces partition the N darts; $\#\{L > T\} \leq N/T$ deterministically.
(3.7)–(3.8) combination	VALID	Every face is light, heavy-short, or long, so $F \leq F_{\text{light}} + F_{\text{heavy,short}} + F_{\text{long}}$. With $\delta \geq cn$: $N/\delta^2 \leq 1/c^2$, $H_N \leq 2 \log n + O(1)$, $N/T = O_c(\sqrt{\log n})$, $Nne^{-\gamma \theta \delta} \leq n^3 \exp(-\gamma cn/\sqrt{\log n}) = o(1)$. All re-derived. The K_n specialization $(2 + o(1)) \log n$ follows ($N/\delta^2 = n/(n-1)$).

step	label	note
(4.1) facial simple-cycle probability	VALID	One constraint per (distinct) vertex, independent rotations, successor of a fixed dart uniform on the other $d - 1$ darts; target $\neq$ source since $v_{i-1} \neq v_{i+1}$ for $\ell \geq 3$. Verified <i>exactly</i> by enumeration: $\Pr = 1/27, 1/81, 1/243$ for oriented 3-, 4-, 5-cycles in K_5 and $1/27, 1/81$ in the octahedron, matching $\prod (d_i - 1)^{-1}$ exactly. Rooting factor $1/\ell$ correct.
(4.2)–(4.3)	VALID	$\text{tr } S^\ell$ telescopes to $\sum_{\text{closed walks}} \prod d_{v_i}^{-1};$
(4.4)	VALID	$1/(d-1) \geq 1/d; \mathbb{E} X_\ell \geq W_\ell/\ell.$ $(S^r)_{vv} = P^r(v, v)$ by similarity via $D^{1/2}; P^r(v, v) \leq \max_u P(u, v) \leq 1/\delta$ using row-stochasticity ($r = 1$ gives 0).
(4.5)	VALID (unused)	Correct ($\text{tr } S^2 = \sum_{uv} A_{uv}/(d_u d_v) \leq n/\delta$) but never invoked afterwards; harmless.
(4.6)–(4.7)	VALID	Walks with $v_i = v_j$ factor into two closed walks at v with multiplying weights; each diagonal entry $\leq 1/\delta$ and nonnegative, sum $\leq n/\delta^2$; length-1 segments vanish (no loops). Union bound over $\binom{\ell}{2}$ position pairs; $W_\ell \geq \text{tr } S^\ell - B_\ell$ holds regardless of trace sign since all discarded weights are ≥ 0 . Verified numerically: $\mathbb{E} X_\ell \geq (\text{tr } S^\ell - B_\ell)/\ell$ on K_5 and octahedron for $\ell = 3, 4, 5$.
(4.8) even/odd pairing	VALID	$\lambda \in [-1, 1]$, each term $\lambda^{2k}(1 + \lambda) \geq 0$, and the eigenvalue 1 (per component, $D^{1/2}\mathbf{1}$) contributes 2. Neatly kills the bipartite -1 issue, and also makes the possible negativity of $\text{tr } S^{2k+1}$ harmless: $\text{tr } S^{2k}/(2k) + \text{tr } S^{2k+1}/(2k+1) \geq (\text{tr } S^{2k} + \text{tr } S^{2k+1})/(2k+1) \geq 2/(2k+1)$ since $\text{tr } S^{2k} \geq 0$.

step	label	note
(4.9)–(4.10)	VALID	$F \geq \sum_{\ell} X_{\ell}$ (distinct lengths, disjoint face counts). Error sum re-derived: $B_{\ell}/\ell = (\ell - 1)n/(2\delta^2)$, total $\leq (n/\delta^2) \sum_{k \leq K} 2k = O(nK^2/\delta^2) \leq 1/16 + o(1)$ with $K = \lfloor \delta/(4\sqrt{n}) \rfloor$. $\sum_{k=2}^K 2/(2k+1) = \log K + O(1) \geq \frac{1}{2} \log n - O_c(1)$ since $\delta \geq cn$. Needs $K \geq 2$, i.e. $n \geq n_0(c)$ — asymptotic statement, fine.
§5 multigraph counterexample	VALID	$F = M - 2g$ (Euler), $F \equiv M \pmod{2}$, so $F \geq 2$ per even dipole; a bridge between components merges exactly one face pair ($F_1 + F_2 - 1$, genus additive). Verified computationally: single dipole $M = 4$ gives only $F \in \{2, 4\}$; chain $k = 3$, $M = 4$: $\min F = 4 = k + 1$ over 20000 samples; chain $n = 12$ ($k = 6$, $M = 12$, $\delta \geq 12$): $\min F = 7 = n/2 + 1$, mean $F \approx 14.1 = \Omega(n)$. So the multigraph version of the conjecture is indeed false and the simplicity hypothesis is essential.

B.3 Reference check

The proof is fully self-contained: it invokes no external theorem (its only “standard” ingredient, the deferred-decision Lemma 2.1, is proved in the writeup and verified here by exhaustive enumeration). Source-paper facts were checked against the arXiv v3 TeX source (downloaded from <https://arxiv.org/eprint/2211.01032v3>):

- **Conjecture 9.3:** confirmed verbatim in Section 9 (\section{Open Problems}), exactly as quoted in prompt.md, third conjecture of the section, with the surrounding context about dense graphs and the CHMMS22 $\Theta(n)$ sparse (multistar, i.e. multigraph) examples.
- **Theorem 1.8** (thm:logBound): confirmed — $\mathbb{E}[F(n)] \leq 3.65 \ln n$ for $n \geq e^{e^{16}}$ (plus $5 \ln n / 10 \ln n + 2$ regimes); lower bound $\frac{1}{2} \ln n - 2$ appears as the separate lower-bound theorem. So the catalog’s “Theorem 1.8 confirms the conjecture for K_n ” is accurate.
- **Mauk–Stahl conjecture** (restated in Section 9): $\mathbb{E}F(K_n) = 2 \ln n + O(1)$ — relevant because the writeup’s K_n specialization $(2 + o(1)) \ln n$ matches this conjectured optimal constant (a strong but not contradictory claim).
- **Model conventions:** the paper defines random embeddings via uniform local rotations of connected graphs — same model as the writeup; “graph” vs “multigraph” usage confirms the simple-graph reading of Conjecture 9.3.

- **Follow-up literature:** two web searches (Sept 2026) found no paper resolving or claiming Conjecture 9.3 for general dense graphs, consistent with the catalog’s four-search audit of 2026-05. Not ALREADY_KNOWN.

B.4 Computational check

Scripts in `verification/scripts/2211.01032__02/` (Python 3, numpy):

- `exact_small.py` — full enumeration of all rotation systems of K_5 (7776) and the octahedron $K_{2,2,2}$ (46656). Results: rooting identity $F = \sum_a 1/L(a)$ holds always; facial-cycle probability (4.1) exact ($1/27, 1/81, 1/243$ vs predictions — equal as rationals); Lemma 3.1 bound holds for $\theta \in \{0.25, 0.5, 0.6\}$ (tight at $\theta = 0.25$ on K_5 : $\max_\ell \Pr = 1/9 = \text{bound}$); lower-bound inequality $\mathbb{E}X_\ell \geq (\text{tr } S^\ell - B_\ell)/\ell$ holds for $\ell = 3, 4, 5$. Exact $\mathbb{E}F(K_5) = 1631/648 \approx 2.517$.
- `lemma21_exposure.py` — all 120 6-cycles, five exposure patterns: conditional image distribution exactly uniform on exactly $d - k - 1$ admissible points.
- `montecarlo.py` — (i) $\mathbb{E}F(K_n)$ for $n = 50 \dots 800$: 6.98, 8.27, 9.52, 11.08, 12.80, i.e. $\approx 2 \ln n - 0.9$ (slope $\approx 2.1 \pm 0.15$ over the range) — consistent with Mauk–Stahl and with the writeup’s upper bound $2 \ln n + O(\sqrt{\log n})$ (empirical values sit *below* $2 \ln n$), and far above the writeup’s lower bound $\frac{1}{2} \ln n$. (ii) $\mathbb{E}F(G(n, 1/2)) \approx 7.8, 8.8, 10.0$ for $n = 100, 200, 400$ — well below the writeup’s $\approx 4 \ln n$ bound. (iii) Unbiased 500k-trace test of Lemma 3.1 on K_{60} , $\theta = 0.3$: $\Pr(L = \ell, \text{light}) \in [2.6, 3.4] \times 10^{-4}$ for $\ell = 3..12$, all below the bound 5.86×10^{-4} , flat in ℓ .
- `dipole_chain.py` — Section 5 counterexample: dipole parity ($F \in \{2, 4\}$ for $M = 4$), chain lower bound $\min F = k + 1$ attained exactly ($k = 3$: min 4; $n = 12$, $k = 6$: min 7, mean 14.1).

No computational check failed; several bounds were found to be exactly tight, matching the writeup’s claims to the constant.

B.5 Caveats

1. **§3.2 off-by-one (the “minor gap”).** “Lemma 2.1 leaves at least $d_u - \lfloor \theta d_u \rfloor$ candidate darts” should be $d_u - \lfloor \theta d_u \rfloor - 1$; the displayed intermediate bound $1/((1 - \theta)\delta)$ in (3.2) is a hair too strong as written, but the only bound used downstream, $2/\delta$, holds for all large n . Routine repair.
2. **Implicit large- n thresholds.** Several steps need $n \geq n_0(c)$ ($\theta\delta \geq 2$, $(\frac{1}{2} - \theta)\delta \geq 1$, $K \geq 2$, $\delta > 1$ for the no-short-faces claim). Harmless for a $\Theta(\log n)$ statement, but the theorem’s “uniformly over all simple graphs” should be read as “for $n \geq n_0(c)$ ”.
3. **Connectivity.** The source’s model is stated for connected graphs; the writeup silently covers disconnected G too (both bounds go through), but never says so.
4. **c -dependent constants.** Upper constant $2/c^2$ blows up as $c \rightarrow 0$; this is inherent to the $\Omega(n)$ hypothesis and consistent with the conjecture’s Θ .
5. **Strength of the byproducts.** If correct, the argument (a) improves the source’s hard-won K_n upper constant from 3.65 to 2, matching the Mauk–Stahl conjectured constant, and (b) improves the source’s $\ln^2 n + 1/p$ bound for dense $G(n, p)$ to $O_p(\log n)$. That a ~4-page elementary argument beats a 55-page SODA paper’s constants is the main reason for residual skepticism; however, the mechanism (light/heavy face split making the folklore $1/\delta^2$ closing heuristic rigorous, plus a spectral count of short simple facial cycles) is exactly the kind of clean idea that can be missed, every step survives detailed scrutiny, and all quantitative predictions match independent numerics.
6. **(4.5) is stated but never used** — cosmetic.
7. The writeup’s own caveat that the multigraph statement is false is correct and computationally confirmed; it does not affect the verdict since the intended reading is simple graphs.

B.6 Referee summary

I approached this as a rejection-seeking referee and failed to find a substantive error. The formalization matches the source’s Conjecture 9.3 under its clearly intended simple-graph reading (verbatim statement confirmed from the arXiv v3 TeX). Every lemma was re-derived in full: the exposure lemma and the facial-cycle probability formula were verified *exactly* by exhaustive enumeration; Lemma 3.1 — the crux of the upper bound, whose conditioning-on- lightness step is the most delicate point — checks out analytically (decomposition over compatible histories) and is exactly tight in small cases; the heavy/short/long face bookkeeping and all arithmetic ($e/25, nK^2/\delta^2 \leq 1/16$, harmonic sums) re-derive correctly; the spectral lower bound handles negative eigenvalues and bipartite components correctly via the even/odd pairing. Monte Carlo at n up to 800 matches all quantitative predictions, including the striking one — $\mathbb{E}F(K_n) \approx 2 \ln n - 0.9$ versus the claimed upper bound $2 \ln n + O(\sqrt{\log n})$. The only literal defects are a one-unit admissible-count slip in §3.2 (conclusion unaffected for large n) and implicit $n \geq n_0(c)$ thresholds, hence MINOR_GAPS rather than CONFIRMED. Because the result would also improve the source paper’s K_n constant to the conjectured Mauk–Stahl optimum, I recommend an independent expert pass before treating the conjecture as resolved, but I could not refute the proof.

References

- [1] J. Champion Loth, K. Halasz, T. Masařík, B. Mohar and R. Šámal, Random embeddings of graphs: the expected number of faces in most graphs is logarithmic, arXiv:2211.01032 (v3, 2025); extended abstract in *Proc. SODA 2024*.